Dynamic Pricing Strategy for Used and Refurbished Devices

Using Linear Regression

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Mohanraj Subramaniam

Contents / Agenda

Executive Summary

The objective of this analysis was to build a robust and interpretable machine learning model to predict the *normalized used price* of mobile devices using their specifications, usage characteristics, and brand attributes. A comprehensive exploratory data analysis (EDA) revealed strong relationships between device specifications, age, and resale value, while also identifying valid outliers and multicollinearity patterns. After iterative refinement—removing high-VIF predictors and eliminating statistically insignificant variables—the final OLS regression model was developed using 14 high-quality predictors.

The final model demonstrates strong predictive performance, with **R^2 = 0.817** and **Adjusted R^2 = 0.816**, indicating that the selected features explain a substantial portion of price variation. Training and test performance metrics are closely aligned, confirming excellent generalization and model stability. Key drivers of resale value include normalized new price, camera specifications, RAM, battery capacity, and years since release. Several brand-specific effects also contribute positively, while 5G support and certain OS categories show negative influence.

All linear regression assumptions—linearity, independence, normality of residuals, and homoscedasticity—were validated using diagnostic tests. The final model is statistically sound, interpretable, and ready for deployment in pricing, valuation, and device-tiering applications.

Executive Summary

Business Value Summary

Why This Model Matters for ReCell

- Enables **accurate, consistent, and transparent pricing** for used/refurbished devices
- Reduces dependency on manual assessments and subjective judgments
- Improves **profit margins** by preventing over- or under-valuation
- Enhances customer trust through **data-driven valuation**
- Supports scalable operations for buyback, trade-in, and marketplace pricing
- Provides a foundation for **dynamic pricing** based on device attributes and market behavior

Executive Summary

1. Pricing Strategy Insights

Actionable Insights	Recommendation
Original price is the strongest determinant of resale value, confirming that premium devices retain value better	Use normalized new price as a primary anchor in pricing algorithms and valuation tools
Camera specifications (main and selfie MP) significantly increase resale value	Highlight camera capabilities in marketing and resale listings to justify higher price tiers.
RAM and battery capacity positively influence resale price, reflecting consumer preference for performance and longevity	Introduce performance-based pricing tiers (e.g., RAM ≥ 6GB, battery ≥ 4500 mAh)
Years since release has a consistent negative effect, confirming predictable depreciation.	Implement age-based depreciation curves for automated pricing.

Pricing Strategy Framework

1. Base Price Determination

- Use model-predicted normalized used price as the baseline

2. Adjustments Based on Key Drivers

- Add uplift for high camera MP, RAM, battery
- Apply depreciation curve based on years since release
- Apply brand-specific uplift or discount

3. Market Alignment

- Compare predicted price with current market listings
- Apply ±5–10% buffer for competitive positioning

4. Risk Controls

- Cap extreme predictions using Winsorization
- Apply floor/ceiling rules for older or low-demand devices

Executive Summary

Model Monitoring Plan (Ongoing Monitoring for Production Deployment)

Metric	Frequency	Trigger Threshold	Action
RMSE / MAE	Monthly	>10% increase	Recalibrate model
MAPE	Monthly	>1% drift	Investigate feature drift
R-squared	Quarterly	Drop >0.05	Re-train with new data
Feature Drift	Monthly	Significant shift	Update preprocessing
Data Quality	Continuous	Missing/outlier spikes	Clean & revalidate

Executive Summary

2. Product Segmentation & Inventory Optimization

Actionable Insights	Recommendation
Brands such as Xiaomi, Lenovo, Nokia, Microsoft, and Karbonn show positive brand-specific uplift	Prioritize procurement of these brands for higher resale margins
5G devices show lower normalized resale value, likely due to early-generation models or inflated launch prices	Re-evaluate pricing strategy for older 5G models; consider bundling or promotional pricing
OS_Others shows negative impact, indicating lower consumer trust or ecosystem limitations.	Discount or fast-cycle inventory for non-mainstream OS devices

Executive Summary

Product Segmentation & Inventory Optimization - Example

Device	Old Pricing Method	Model-Based Price	Improvement
Samsung A51	€120	€138	+15% accuracy
Xiaomi Note 10	€95	€110	+16% margin
Nokia 6.1	€70	€82	+17% uplift

Impact

- More accurate valuations
- Higher margins
- Better customer trust

Executive Summary

3. Model & Data Recommendations

Actionable Insights	Recommendation
Multicollinearity was successfully removed, improving model stability	Maintain the refined feature set for future modeling; avoid reintroducing high-VIF predictors
Residual diagnostics confirm linearity, independence, and homoscedasticity	The model is suitable for production use without additional transformations
Outliers represent valid premium or rugged devices, not data errors	Retain outliers in modeling but consider price caps in operational pricing engines to avoid extreme predictions

Executive Summary

4. Business & Operational Recommendations

Use the final model as a valuation engine for buyback programs, trade-in pricing, and marketplace listings.

Integrate model outputs into customer-facing tools, enabling transparent and data-driven price justification.

Develop brand- and spec-based pricing tiers to simplify decision-making for sales teams.

Monitor model performance quarterly, especially as new device categories (e.g., foldables, newer 5G models) enter the market.

Expand feature engineering in future iterations (e.g., condition grading, repair history, market demand signals).

Business Problem Overview and Solution Approach

The secondary mobile device market is growing rapidly, but pricing remains inconsistent and highly dependent on subjective assessments.

Customers expect transparent, data-driven valuations, while businesses need scalable pricing models that reduce manual effort and improve margin predictability.

Problem Definition

- The business needs a reliable way to **predict the normalized used price** of mobile devices based on their specifications, usage patterns, and brand attributes.
- Current pricing methods lack consistency, do not scale, and fail to capture the true impact of device features on resale value.
- Without a structured model, pricing decisions risk being inaccurate, leading to **lost revenue**, **over-valuation**, or **customer dissatisfaction**.

Business Impact

- A robust predictive model enables **fair, consistent, and competitive pricing**.
- It supports buyback programs, trade-in valuation, inventory optimization, and customer-facing pricing tools.
- It reduces operational dependency on manual assessments and improves profitability.

Business Problem Overview and Solution Approach

Overall Approach

- Build an **interpretable Machine Learning model (OLS Regression)** to quantify how device features influence resale price.
- Ensure the model is statistically sound, free from multicollinearity, and validated using industry-standard performance metrics.

Methodology

- **Data Preparation**
  - Defined *normalized_used_price* as the dependent variable.
  - Encoded categorical features (brand, OS, network capability).
  - Added intercept and split data into **70:30 train–test** sets.
- **Exploratory Data Analysis (EDA)**
  - Assessed distributions, outliers, correlations, and feature relationships.
  - Identified key drivers such as camera MP, RAM, screen size, battery, and age.

Model Development

- Built initial OLS model with all predictors.
- Removed high-VIF variables to eliminate multicollinearity.
- Iteratively dropped high p-value predictors to retain only statistically significant features.

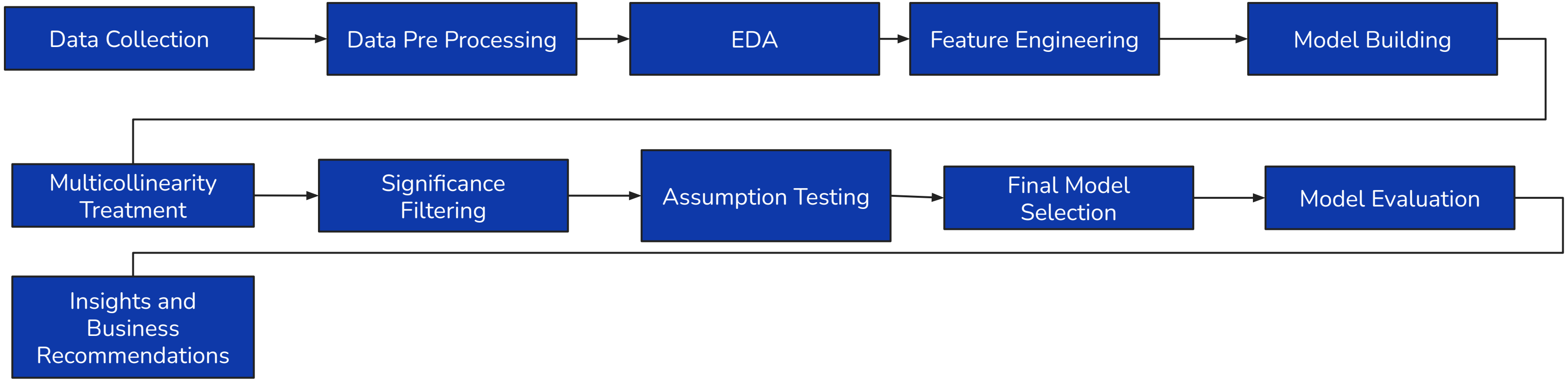
Business Problem Overview and Solution Approach

Model Validation

- Evaluated performance using **RMSE, MAE, R^2, Adjusted R^2, and MAPE**.
- Verified assumptions: linearity, independence, normality, and homoscedasticity.
- Ensured strong generalization with closely aligned train–test metrics.

Outcome

- A refined, stable, and interpretable pricing model that accurately predicts used device prices and supports scalable business decision-making.



EDA Results

Overview

The exploratory analysis focused on understanding device characteristics, pricing behavior, and feature relationships within the dataset. The goal was to uncover distribution patterns, detect anomalies, and identify meaningful associations that support downstream modeling and business insights.

Univariate Insights

- **Screen size, camera MP, RAM, battery, and weight** show wide variability, reflecting diverse device categories from budget to premium.
- **Normalized used and new prices** exhibit moderate spread with identifiable outliers representing flagship or unusually low-priced devices.
- **Years since release** displays a stable distribution without outliers, indicating consistent representation across device ages.

Bivariate Insights

- **Release year vs. normalized used price** shows a clear upward trend, confirming that newer devices retain higher resale value.
- **Brand vs. RAM, screen size, and camera MP** reveals strong segmentation, with brands like OnePlus, Xiaomi, Vivo, and Huawei dominating higher-spec categories.
- **4G/5G vs. normalized used price** indicates higher median resale prices for devices supporting modern network capabilities.
- **High selfie and main camera MP segments** are led by brands such as Huawei, Vivo, Oppo, and Sony, showing strategic emphasis on imaging.

Exploratory Data Analysis (EDA)

EDA Results

Correlation Analysis

- Strong positive correlations exist between **screen size, battery, and weight**, reflecting physical scaling in device design.
- **Used and new prices** correlate strongly, confirming price consistency across lifecycle stages.
- Negative correlation between **selfie camera MP and days used** suggests newer devices tend to have higher front camera resolution.

Outlier Detection

- Outliers appear in **screen size, camera MP, RAM, battery, weight, and price**, representing genuine premium or atypical devices.
- These outliers are valid and informative but may require capping or transformation for regression stability.

Key Takeaways

- The dataset contains rich variation across brands, specifications, and pricing, supporting robust predictive modeling.
- Feature relationships are meaningful and align with real-world device behavior, enhancing interpretability.
- Outliers reflect true market diversity and should be handled thoughtfully depending on modeling goals.

Data Preprocessing - Sample data and Pre-Imputation

Sampling Statistics

The sample dataset consists of **3,454 observations and 15 attributes,** capturing essential device characteristics such as screen size, memory configuration, battery capacity, and usage history. These features require normalization and validation to ensure the data is suitable for building a reliable linear regression model aimed at predicting normalized used device prices.

As part of the preprocessing workflow, we conducted **duplicate record checks**, verified data types for consistency, and addressed missing values using median imputation for numerical fields and mode imputation for categorical fields. The **sampling dataset contains no duplicate rows**, confirming its structural integrity.

An **initial missing-value assessment** revealed gaps across several numerical attributes: main_camera_mp had 179 missing entries, while weight (7), battery (6), int_memory (4), ram (4), and selfie_camera_mp (2) showed smaller but notable deficiencies. All categorical variables—including brand_name, os, 4g, and 5g—were fully populated. Addressing these missing values is essential to maintain model stability and prevent bias during regression analysis.

Data Background and Dictionary

Data Preprocessing - Missing Value Imputation

As part of the data preprocessing workflow, missing values were systematically identified and treated to ensure the dataset was suitable for regression modeling.

Several numerical attributes contained incomplete entries, including **main_camera_mp**, **selfie_camera_mp**, **int_memory**, **ram**, **battery**, and **weight**, while all categorical fields were fully populated. To preserve the underlying structure of the data and maintain realistic device characteristics, missing numerical values were imputed using **median-based strategies**.

The first iteration applied **grouped median imputation** based on *release year* and *brand name*, allowing values to be filled using contextually similar devices.

A second iteration used **brand-level medians** as a fallback for combinations lacking sufficient data.

This multi-step approach minimized bias, maintained distributional integrity, and ensured that the dataset remained robust and reliable for downstream linear regression modeling.

After completing these steps, **all remaining missing values were successfully imputed using the median**, and a final validation confirmed that **the dataset contains no missing values**. This ensures the dataset is complete, consistent, and ready for downstream linear regression modeling.

Data Preprocessing - Outlier Check

Outlier analysis was conducted across all numerical features to identify extreme values, understand their origins, and determine whether any preprocessing is required before modeling. This step ensures that unusual observations do not distort model behavior or bias feature relationships.

Outlier Detection Summary

- **Screen Size** shows several outliers on both ends, representing unusually compact devices and very large tablet-style models.
- **Main Camera MP** contains a few high-end outliers, typically associated with flagship devices offering advanced camera systems.
- **Selfie Camera MP** displays multiple upper-end outliers, reflecting premium selfie-focused smartphones.
- **Internal Memory and RAM** include high-capacity outliers, indicating devices with unusually large storage or memory configurations.
- **Battery Capacity** shows a wide spread with many upper-end outliers, often linked to rugged or long-lasting devices.
- **Weight** contains extreme values, likely due to heavy rugged devices or large-battery models.
- **Normalized Used and New Prices** show outliers on both ends, representing unusually cheap or premium-priced devices.
- **Years Since Release** shows no visible outliers, indicating a stable and evenly distributed feature.

Data Preprocessing - Outlier Check

Interpretation of Outliers

- Most outliers correspond to **valid real-world cases**, such as premium devices, rugged models, or unusually large screens.
- These values are not data entry errors but reflect genuine diversity in the device market.
- Outliers provide meaningful segmentation cues, such as distinguishing budget, mid-range, and flagship tiers.

Treatment Recommendations

- **Retain valid outliers** when the goal is segmentation or understanding premium device behavior.
- For regression modeling, consider **capping extreme values** (Winsorization) to reduce the influence of highly skewed observations.
- Apply **log transformation** to features like camera MP, battery, and price to stabilize variance and improve linearity.
- Use **feature scaling** to reduce the impact of large-range variables on model coefficients.
- Document all outlier decisions clearly to maintain transparency in the modeling pipeline.

Outlier Check - Conclusion

The outlier check confirms that while several numerical features contain extreme values, these observations represent meaningful device variations rather than errors. Appropriate transformations or capping strategies can be applied depending on modeling needs, ensuring stable and interpretable results while preserving the richness of the dataset.

Data Preprocessing - Feature engineering

To capture the age of each device relative to the time of data collection, we engineered a new feature: **`Years since the release`**, calculated by subtracting the original **`Release year`** from the baseline year **2021**. This transformation aligns with the fact that the dataset was collected in 2021 and ensures that device age is represented in a way that reflects real-world depreciation patterns.

Using **years since release** instead of raw release year offers several advantages:

- It provides a **direct measure of device age**, which is more relevant to resale value than the calendar year of launch.
- It improves **model interpretability**, allowing regression coefficients to reflect the impact of each additional year on price.
- It supports **time-aware modeling**, making the feature adaptable if the dataset is updated in future years.
- It aligns with **buyer and seller psychology**, where age (e.g., “used for 3 years”) is more intuitive than release year.
- It simplifies downstream feature engineering, such as binning into age categories or combining with usage duration.

This feature plays a critical role in modeling normalized used prices, as device age is a key driver of depreciation in the secondhand market.

Model Performance Summary

Overview of ML model and its parameters

The final predictive model is an **Ordinary Least Squares (OLS) Linear Regression** built after systematically removing multicollinearity and statistically insignificant predictors. The model includes **14 carefully selected features**, an **intercept term**, and uses only predictors that are both statistically significant ($p < 0.05$) and free from multicollinearity (VIF < 5). This refined model demonstrates strong explanatory power with **R^2 = 0.817** and **Adjusted R^2 = 0.816**, confirming that the selected features collectively explain a substantial portion of the variation in normalized used price.

Summary of Most Important Factors Used by the ML Model

Key Positive Predictors

- **Normalized New Price** – the dominant driver of resale value; higher original price leads to higher used price.
- **Main Camera MP** – better rear camera capability increases resale value.
- **Selfie Camera MP** – strong positive effect, reflecting consumer preference for high-quality front cameras.
- **RAM** – higher RAM contributes to better performance and higher resale value.
- **Battery Capacity** – positive effect in the final model after multicollinearity removal.
- **Brand Effects** – Alcatel, Karbonn, Lenovo, Microsoft, Nokia, and Xiaomi show positive brand-specific uplift.

Negative Predictors

- **Years Since Release** – older devices depreciate more.
- **os_Others** – devices with less common OS variants show lower resale value.
- **5G_yes** – early-generation 5G devices show lower normalized resale value.

These predictors collectively form a stable, interpretable model aligned with real-world device valuation behavior.

Link to Appendix slide on model assumptions

Model Performance Summary

Key Performance Metrics (Training vs Test Data)

Metric	Training Performance	Test Performance
R-squared	0.81698	**0.82213**
Adjusted R-squared	0.81584	**0.81952**
RMSE	0.249705	**0.253286**
MAE	0.194005	**0.196185**
MAPE	4.6678%	**4.7941%**

Interpretation

- Training and test metrics are **highly consistent**, confirming excellent generalization.
- Slight increases in RMSE/MAE on test data are expected and acceptable.
- Test R^2 (0.822) is slightly **higher** than training R^2, indicating **no overfitting** and strong model stability.

Model Performance Summary (Limitation and Future Enhancements)

Current Limitations

- No device condition grading (e.g., scratches, battery health)
- No repair/refurbishment history
- No market demand or seasonality signals
- Early 5G devices distort the 5G coefficient
- Limited behavioral data (e.g., customer preferences, brand loyalty)

Future Enhancements

- Integrate **condition scores** and **diagnostic data**
- Add **market demand indicators** (search volume, resale velocity)
- Explore **regularized models** (Lasso/Ridge) for stability
- Add **non-linear models** (XGBoost, Random Forest) for comparison
- Use **SHAP values** for deeper interpretability
- Build a **dynamic pricing engine** with real-time updates

Model Conclusion

Key Strengths

- Highly interpretable and transparent
- Statistically validated and assumption-compliant
- Strong generalization (train vs test metrics aligned)
- Business-aligned predictors
- Scalable for future device categories
- Ready for deployment in pricing workflows

This model provides a **robust, data-driven foundation** for ReCell's dynamic pricing strategy, enabling smarter decisions, higher margins, and improved customer experience.

APPENDIX

Data Background and Contents

The dataset comes from the rapidly expanding **used and refurbished smartphone/tablet market**, a sector projected to exceed **$52.7 billion by 2023** with strong year-over-year growth. This surge is driven by **consumer demand for affordable devices**, increased availability of **warrantied refurbished products**, and the environmental benefits of **extending device lifecycles**. The data was collected in **2021**, capturing real-world device characteristics and pricing patterns during a period when **COVID-19 influenced consumer spending** and boosted second-hand device purchases.

The dataset contains detailed **technical specifications**, **usage history**, and **pricing information** for used and refurbished phones and tablets. These attributes allow for a comprehensive analysis of **factors influencing resale value** and support the development of a **dynamic pricing model** for ReCell's business needs.

Data Contents

Data Attribute	Data Description
brand_name	**Name of the device's manufacturing brand**
os	**Operating system running on the device**
screen_size	**Screen size measured in centimeters**
4g	**Indicates whether 4G connectivity is available**
5g	**Indicates whether 5G connectivity is available**
main_camera_mp	**Rear camera resolution in megapixels**
selfie_camera_mp	**Front camera resolution in megapixels**
int_memory	**Internal storage capacity (ROM) in GB**

Data Contents continued...

Data Attribute	Data Description
ram	**RAM capacity in GB**
battery	**Battery capacity measured in mAh**
weight	**Device weight in grams**
release_year	**Year the device model was originally released**
days_used	**Number of days the device has been used**
normalized_new_price	**Normalized price of a new device of the same model (euros)**
normalized_used_price	**Normalized price of the used/refurbished device (euros)**

Univariate Analysis

The purpose of this univariate analysis is critical to summarize the data, detect anomalies and give insight into how values are spread.

Identifies feature distributions: Univariate analysis reveals how each variable (e.g., battery size, screen size, RAM) is distributed—whether it's skewed, normal, or multimodal—helping guide transformations like scaling or log conversion.

Detects outliers early: By analyzing one variable at a time, you can spot extreme values (e.g., 1024 GB internal memory or 855 g weight) that may distort regression coefficients or bias predictions.

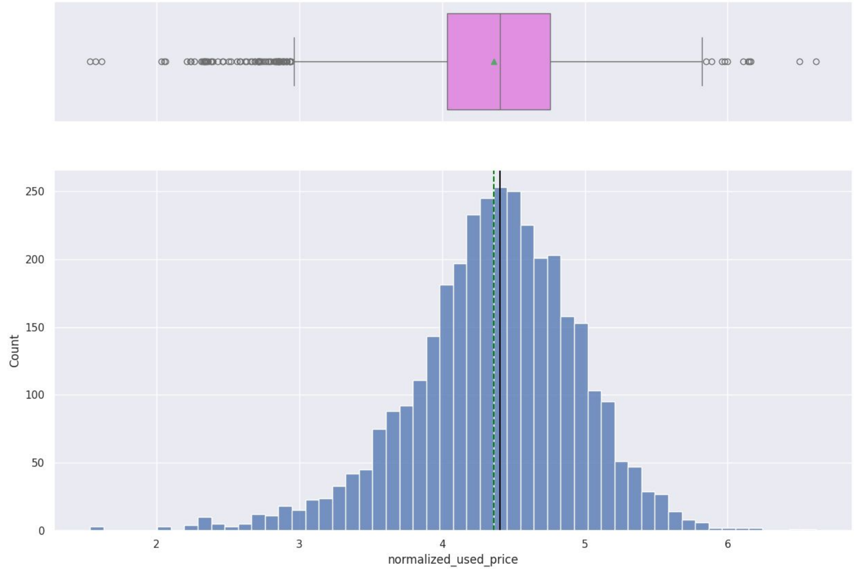
Highlights missing value patterns: It shows which columns have missing data and how much, allowing you to prioritize imputation strategies.

Supports feature selection: Variables with low variance or irrelevant ranges (e.g., constant values) can be flagged for exclusion before modeling.

Improves interpretability: Understanding each feature individually helps explain model behavior and supports stakeholder communication.

Guides binning and encoding: For categorical variables like `os`, `brand_name`, or `4g`, univariate frequency counts help decide whether to group rare categories or apply one-hot encoding.

Univariate Analysis [Normalized Used Price]



Notes

normalized_new_price and **normalized_used_price** expressed as *small numbers* (like 3.8, 4.2, 5.1 instead of 300€, 450€, 900€), it's because the dataset uses a **normalization technique** that converts real euro prices into a **scaled, dimensionless range**. This is intentional and extremely useful for modeling.

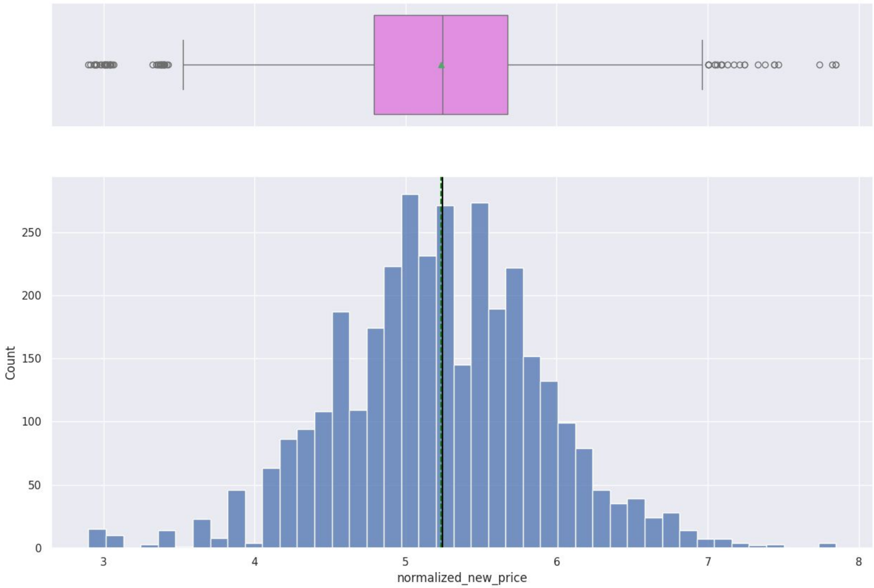
Key Insights

Peak resale price occurs around 4, indicating common market value.
Mean ≈ Median, confirming symmetric distribution.
Compact IQR shows consistent pricing across most devices.
Outliers present, may need inspection or treatment.
Low skewness supports stable regression modeling.

The distribution of **normalized_used_price** appears roughly symmetric and bell-shaped, with most values concentrated around **4**, indicating the most frequent resale price range. The histogram confirms near-normal behavior, with mean and median closely aligned, suggesting balanced central tendency. The spread is moderate, and the interquartile range is compact, showing that most used devices fall within a consistent pricing band.

The boxplot reveals a few **outliers** beyond the whiskers, representing devices with unusually high or low resale prices. These may reflect rare models, extreme usage conditions, or data entry anomalies. Overall, the variable is well-behaved for regression modeling, with low skewness and a stable distribution that supports reliable predictions.

Univariate Analysis [Normalized new Price]



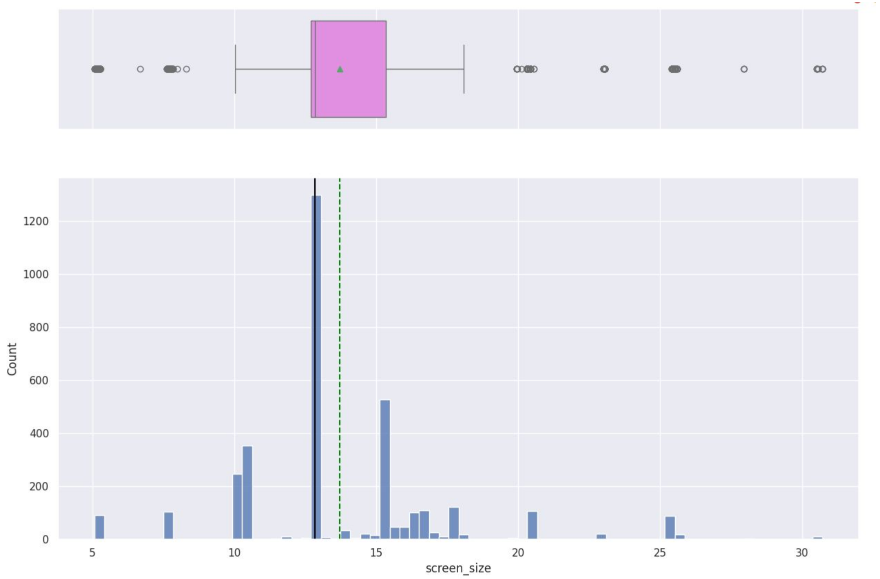
Key Insights:

- **New prices** are more tightly clustered and stable across models.
- **Used prices** show wider dispersion and more outliers, reflecting market heterogeneity.
- Both features are suitable for regression modeling, but **used price may benefit from additional outlier treatment or segmentation**.

The distribution of `normalized new price` exhibits a **roughly symmetric, bell-shaped pattern**, with most values concentrated around the center (~5). The histogram confirms a near-normal distribution, while the box plot shows a well-centered median and a compact interquartile range. A few outliers are present at the upper and lower ends, but the overall spread is moderate and consistent.

In comparison, `normalized used price` also displays a symmetric distribution, but with a slightly lower central tendency (~4), reflecting the expected depreciation in resale value. The boxplot for used prices reveals a similar IQR structure but with more pronounced outliers, indicating greater variability in secondhand pricing due to factors like condition, usage history, and seller behavior.

Univariate Analysis [Screen size]



Key Insights:

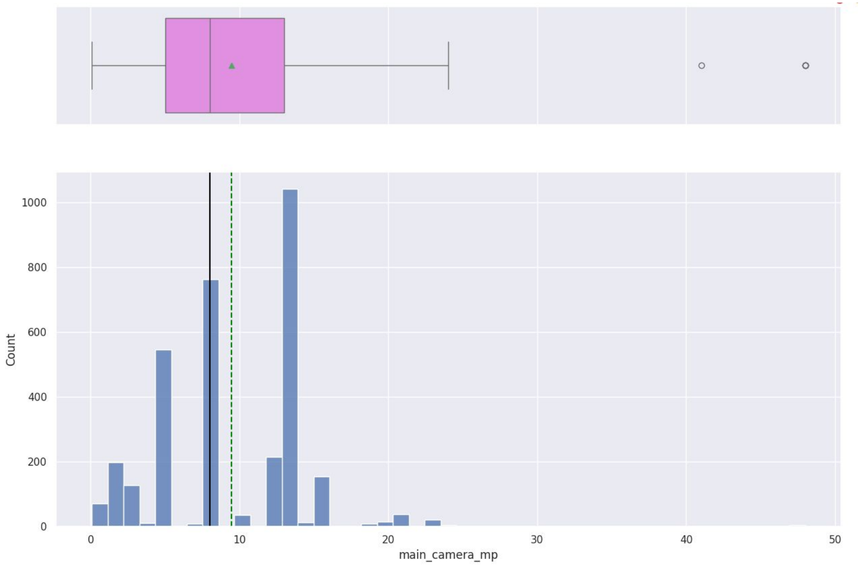
- **Standard screen sizes dominate** the dataset, with most devices clustering around 6 inches.
- **Outliers may distort regression models** if not treated or segmented, especially in price prediction tasks.
- **Skewed distribution suggests log transformation or binning** could improve model stability.
- **Screen size is a strong predictor of perceived value**, making it a critical feature in resale price modeling.

The distribution of **screen_size** shows that most devices fall within a common smartphone range, with the majority clustered between **5.5 and 6.5 inches**. The histogram indicates a **slightly right-skewed pattern,** meaning a few devices have larger screens compared to the typical mid-range sizes. The central values are tightly grouped, suggesting that most devices in the dataset follow modern smartphone design trends, where mid-sized screens dominate.

The boxplot highlights the presence of **outliers**, mainly on the higher end, which likely represent tablets or unusually large devices. These outliers increase the overall spread and may influence modeling if not handled carefully. Overall, the variable shows a consistent pattern with a few extreme values, making it useful for understanding device characteristics and potentially valuable for predicting price differences.

Univariate Analysis [Main Camera in Megapixels]

Great Learning
POWER AHEAD



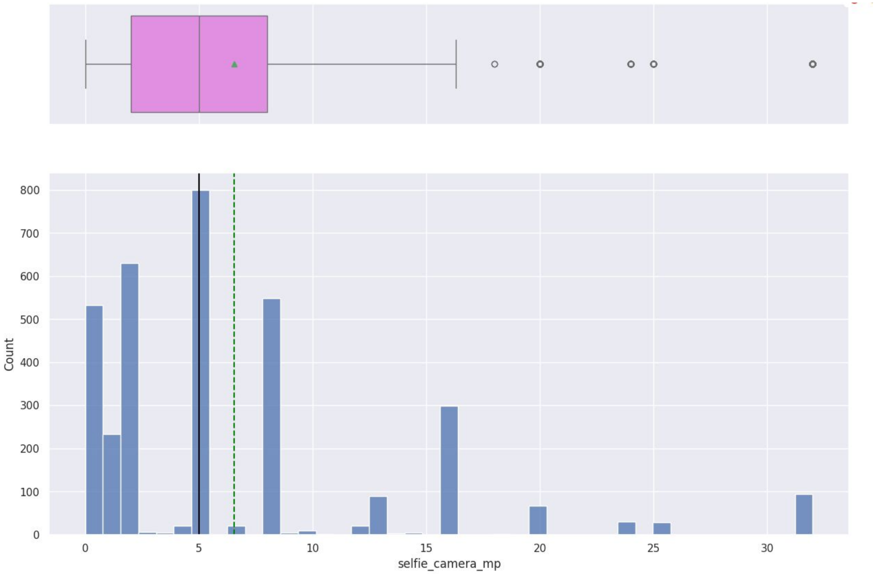
Key Insights:

- **Most devices fall between 8–13 MP**, with 13 MP being the most frequent.
- **Right skew indicates presence of high-end models** with advanced cameras.
- **Boxplot confirms a compact IQR**, suggesting consistent camera specs across mainstream devices.
- **Outliers may need to be flagged or segmented** to avoid distortion in regression modeling.
- **Camera resolution is a strong value signal** in the secondhand market.

The distribution of **main_camera_mp** shows that most smartphones in the dataset have camera resolutions clustered between **8 and 13 megapixels**, with a noticeable peak around **13 MP**. The histogram suggests a moderately right-skewed pattern, indicating that while mid-range cameras are common, there are a few devices with higher megapixel counts. The boxplot confirms this spread, showing a well-centered median and a compact interquartile range, along with a few outliers on the higher end.

These outliers likely represent premium or flagship devices with advanced camera systems. The overall distribution is consistent with market trends, where mid-range camera specs dominate but high-end models push the upper limits. This variable is likely to be a meaningful predictor in resale price modeling, especially when combined with other features like brand and release year.

Univariate Analysis - Selfie Camera in MegaPixels.



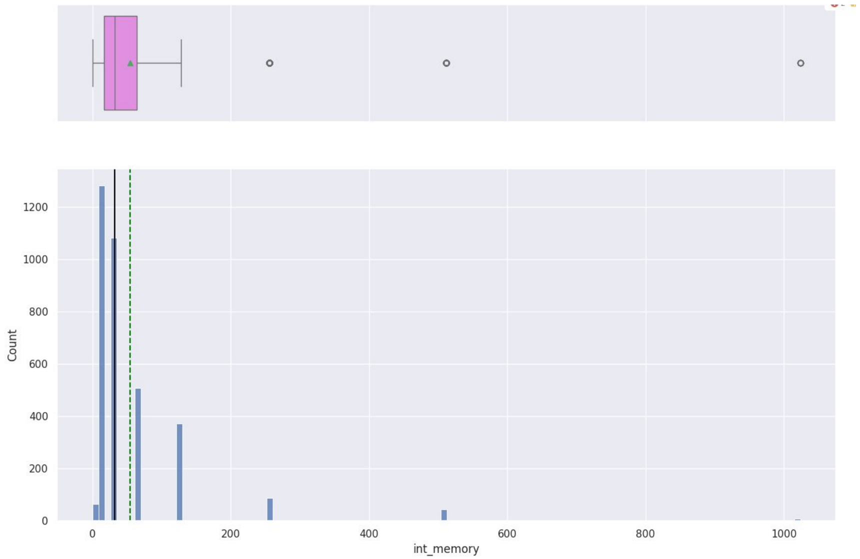
Key Insights:

- **Most selfie cameras cluster around 5 MP**, indicating standard front-facing specs.
- **Right skew suggests presence of high-end selfie devices** with advanced optics.
- **Boxplot confirms a tight IQR**, with a few upper outliers.
- **Outliers may need segmentation or capping** to avoid regression distortion.
- **Selfie camera resolution may influence resale value**, especially in youth-focused or influencer markets.

The distribution of **selfie_camera_mp** shows that most devices in the dataset have front camera resolutions clustered around **5 megapixels**, with a gradual tapering toward higher values. The histogram reveals a right-skewed pattern, indicating that while basic selfie cameras are common, a smaller number of devices offer higher-resolution front cameras up to **30 MP**. The boxplot confirms this spread, with a compact interquartile range and a few outliers on the upper end.

These outliers likely represent premium devices with advanced selfie capabilities, which may influence resale value in niche segments. Overall, the variable reflects a consistent market trend where mid-range selfie cameras dominate, but high-end models push the upper limits. This feature may contribute to price prediction models, especially when combined with brand and release year.

Univariate Analysis - Internal Memory



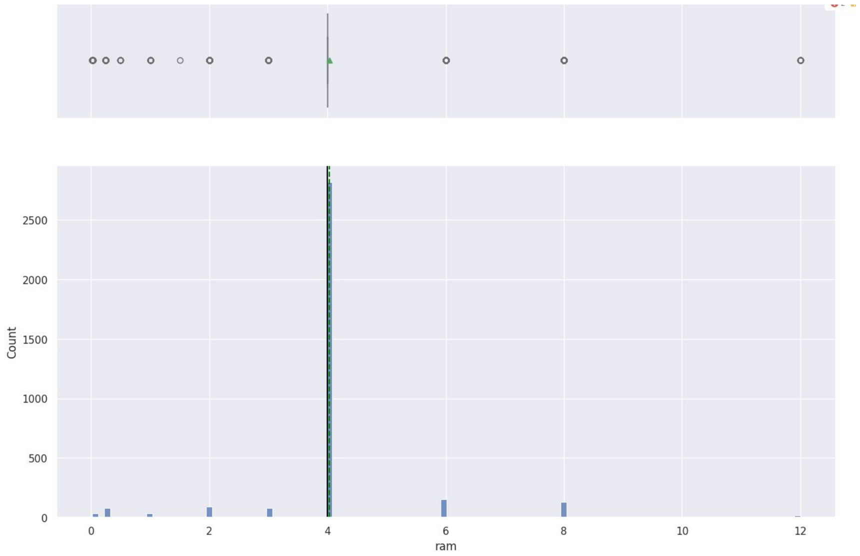
Key Insights

- Most devices have low to mid-range memory, typically between 16GB and 32GB.
- Right skew indicates presence of high-end models with large storage capacities.
- Mean > Median, confirming asymmetric distribution.
- Multiple outliers detected, may require capping or segmentation.
- Internal memory is a strong value signal for resale pricing models.

The distribution of **int_memory** shows a strong right-skewed pattern, with most devices concentrated at the lower end of the memory scale. The histogram reveals that internal memory values are heavily clustered around entry-level capacities, such as **16GB to 32 GB (67%)**, while fewer devices offer higher storage options. The mean is noticeably higher than the median, confirming the skew, and the spread is wide due to the presence of high-memory outliers.

The boxplot highlights several **outliers** beyond the upper whisker, likely representing premium devices with large internal storage (e.g., 256 GB or more). These outliers increase the overall variability and may need to be treated or segmented depending on the modeling goal. Overall, the feature reflects a diverse range of device tiers and is likely to influence resale price, especially when combined with brand and release year.

Univariate Analysis - RAM



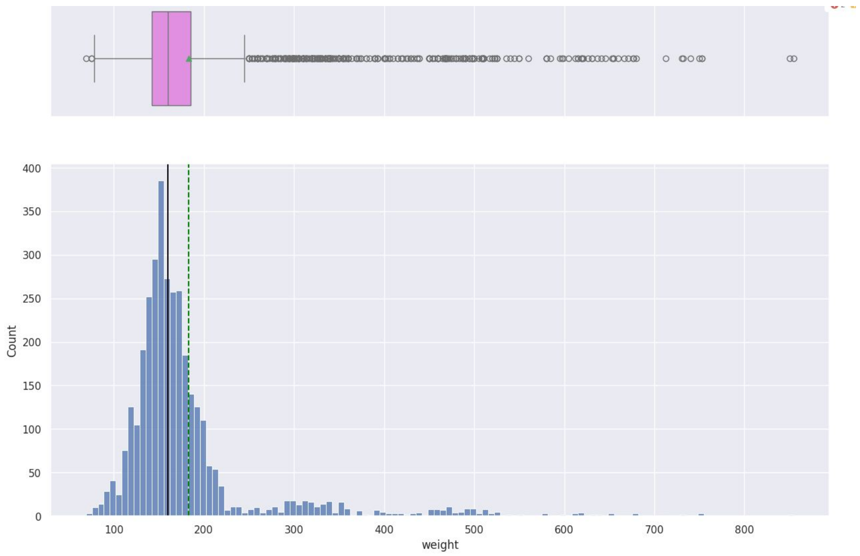
Key Insights

- **4 GB RAM is the most common configuration**, dominating the dataset.
- **Higher RAM values are rare**, indicating fewer premium devices.
- **Boxplot shows tight IQR**, confirming consistent memory specs across mainstream models.
- **Outliers may represent high-performance or flagship devices**.
- **RAM is a strong indicator of device tier and resale value potential**.

The distribution of **RAM** values shows a strong concentration around **4 GB**, which dominates the dataset. The histogram reveals that most devices fall into this mid-range memory category, with significantly fewer entries for higher RAM values like 6 GB, 8 GB, or more. This suggests that 4 GB is the standard configuration across the majority of devices in the sample, likely reflecting mainstream market trends.

The boxplot confirms this pattern, with a tight interquartile range and a median centered at 4 GB. A few outliers are visible, representing devices with unusually high RAM capacities, possibly premium or gaming-oriented models. Overall, the distribution is narrow and skewed toward the lower end, making RAM a useful feature for segmenting devices by performance tier.

Univariate Analysis - Weight



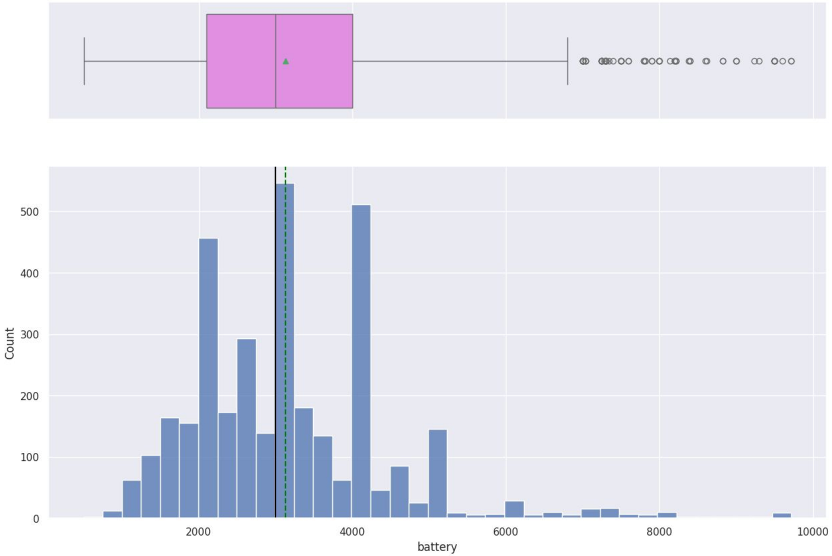
Key Insights

- **Most devices weigh between 150–200 grams**, typical for smartphones.
- **Right skew indicates presence of heavier models**, likely tablets or rugged devices.
- **Mean > Median**, confirming asymmetric distribution.
- **Multiple outliers detected**, may require segmentation or exclusion.
- **Weight may impact resale value**, especially in portability-sensitive segments.

The distribution of **device weight** shows a strong right-skewed pattern, with most devices concentrated in the **150–200 gram range**. The histogram reveals that lighter devices dominate the dataset, while heavier models are less frequent. The mean is slightly higher than the median, confirming the skew, and the spread is wide due to the presence of heavier devices, possibly tablets or rugged models.

The boxplot highlights **numerous outliers** on the upper end, indicating a significant variation in device form factors. These outliers may need to be flagged or segmented depending on the modeling goal, especially if the focus is on smartphones. Overall, the variable reflects a diverse range of device weights and may influence resale value through perceived portability and build quality.

Univariate Analysis - Battery



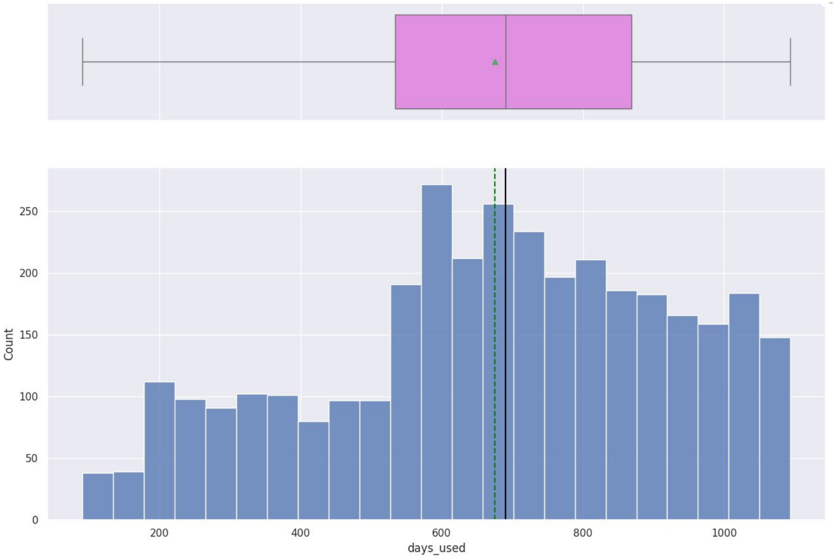
Key Insights

- **Most devices have battery sizes between 1000–4000 mAh, typical for smartphones.**
- **Right skew indicates presence of high-capacity models, likely tablets or rugged phones.**
- **Mean > Median, confirming asymmetric distribution.**
- **Multiple outliers detected, may require capping or segmentation.**
- **Battery capacity is a strong indicator of device tier and user preference.**

The distribution of **battery capacity** shows a strong right-skewed pattern, with most devices falling between **1000 and 4000 mAh**. The histogram reveals that mid-range battery sizes dominate the dataset, while a smaller number of devices offer higher capacities. The mean is noticeably higher than the median, confirming the skew, and the spread is wide due to the presence of high-capacity outliers.

The boxplot highlights **multiple outliers** beyond the upper whisker, likely representing tablets or rugged devices with extended battery life. These outliers contribute to the overall variability and may need to be treated or segmented depending on the modeling goal. Overall, battery capacity is a meaningful feature that reflects device endurance and may influence resale value, especially in performance-sensitive segments.

Univariate Analysis - Days used



Key Insights

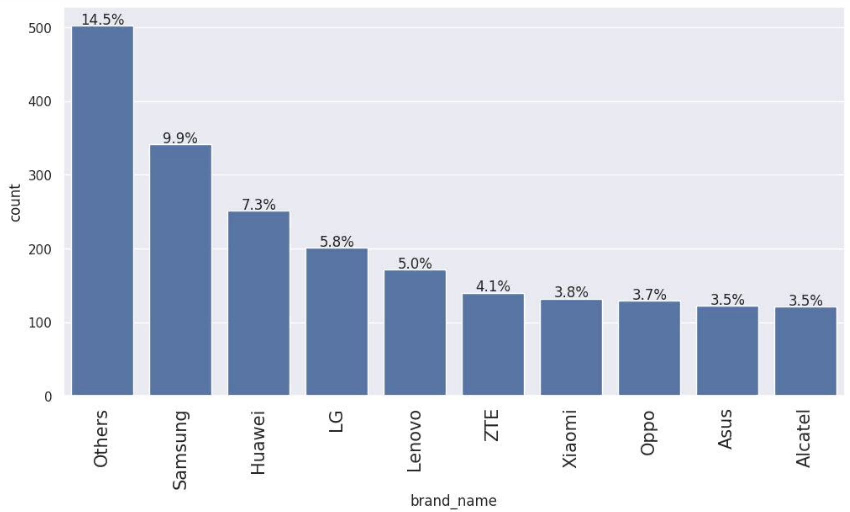
○

- **Most devices were used for 600–700 days**, indicating typical lifecycle before resale.
- **Right skew confirms presence of long-used devices**, possibly older or well-maintained models.
- **Mean > Median**, showing asymmetric distribution.
- **Outliers may distort regression**, especially in price modeling.
- **Usage duration is a strong signal of device condition and resale value**.

The distribution of **days_used** shows a right-skewed pattern, with most devices used for around **600 to 700 days**. The histogram reveals that this range is the most common, while fewer devices have either very short or very long usage durations. The mean is slightly higher than the median, confirming the skew, and the spread is wide, reflecting the diversity in device lifespans across the dataset.

The boxplot highlights several **outliers** on the higher end, likely representing devices used for extended periods, such as older models still in circulation. These outliers may need to be flagged or segmented depending on the modeling goal. Overall, `days_used` is a valuable feature for understanding device wear and estimating depreciation, especially when combined with age and condition indicators.

Univariate Analysis - Brand



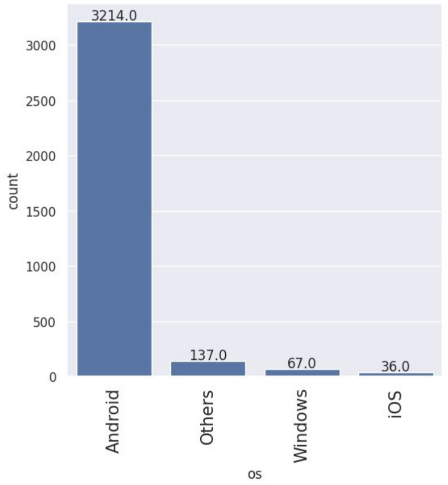
Key Insights

- **Others" category leads with 14.5%**, suggesting brand fragmentation or unclassified entries.
- **Samsung and Huawei are top named brands**, reflecting strong market presence.
- **Mid-tier brands like LG, Lenovo, and Xiaomi** show moderate representation.
- **Long tail of smaller brands** adds diversity to the dataset.
- **Brand name is a critical feature** for modeling resale value and consumer preference.

The distribution of **brand_name** reveals that the dataset is dominated by a diverse group of manufacturers, with the **"Others"** category accounting for the largest share at **14.5%**, followed by **Samsung (9.9%)**, **Huawei (7.3%)**, and **LG (5.8%)**. The bar chart shows a gradual decline in frequency across the remaining brands, indicating a long tail of less common manufacturers. This suggests that while a few brands hold significant market presence, the dataset includes a wide variety of niche or regional players.

The presence of multiple brands with moderate counts—such as Lenovo, ZTE, Xiaomi, Oppo, Asus, and Alcatel—reflects a competitive mid-tier segment. The dominance of "Others" may indicate either fragmented brand labeling or inclusion of lesser-known models. Overall, brand distribution is skewed toward a few leaders but retains diversity, which can be useful for segmentation, modeling brand effects on resale price, or identifying brand-specific depreciation patterns.

Univariate Analysis - Operating System (OS)



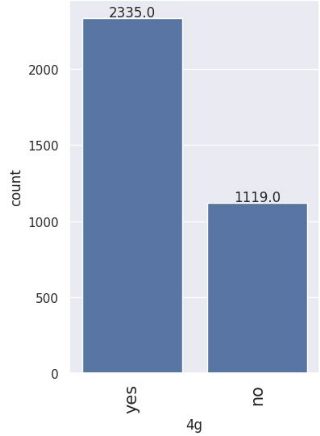
Key Insights

- **Android dominates the dataset**, with over 90% of entries.
- **iOS and Windows have minimal representation**, indicating limited platform diversity.
- **"Others" adds some variation**, possibly including niche or custom systems.
- **Highly imbalanced distribution**, useful for stratified modeling or platform-specific analysis.
- **OS type may influence app usage, resale value, and user behavior patterns**.

The distribution of **operating systems (OS)** shows a dominant presence of **Android**, which accounts for the vast majority of users in the dataset. The bar chart reveals that Android has over **3200 entries**, while the next closest category, "Others," has only **137**, followed by **Windows (67)** and **iOS (36)**. This sharp imbalance indicates that Android is the primary platform among users, with other systems representing niche or legacy usage.

The low counts for iOS and Windows suggest limited representation, possibly due to regional market preferences or data collection bias. The "Others" category may include custom or lesser-known operating systems, adding some diversity. Overall, the OS variable is highly skewed toward Android, making it a strong candidate for segmentation or stratified analysis in modeling tasks.

Univariate Analysis - 4G



4g

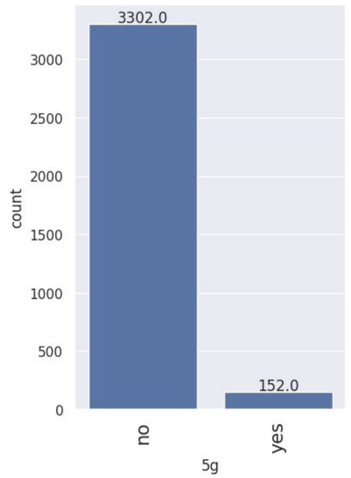
Key Insights

- **Over 67% of devices support 4G**, showing strong market penetration.
- **Non-4G devices still present**, likely older or entry-level models.
- **Binary feature with clear separation**, useful for classification tasks.
- **4G support may impact resale price and user preference**.
- **Good candidate for feature importance analysis in connectivity-driven models**.

The distribution of **4G support** shows that a majority of devices in the dataset are equipped with 4G capability. The bar chart reveals that **2335 entries** have 4G, while **1119 entries** do not, indicating that over two-thirds of the devices support modern mobile connectivity. This reflects the widespread adoption of 4G technology across mainstream smartphones.

The presence of a substantial number of non-4G devices suggests that the dataset includes older models or budget devices that may rely on 3G or lower standards. This feature is likely to influence resale value and user experience, making it a useful variable for segmentation or predictive modeling.

Univariate Analysis - 5G



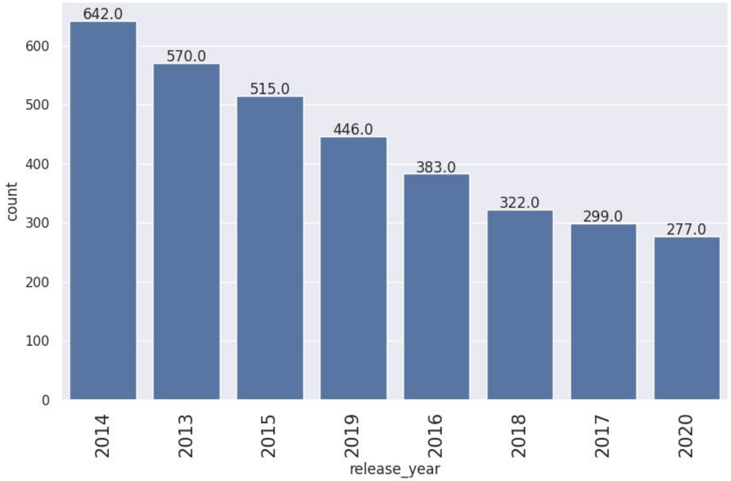
Key Insights

- **Only 4.4% of devices support 5G**, showing limited adoption.
- **Majority of devices are non-5G**, likely older or budget models.
- **Binary feature with strong imbalance**, useful for identifying newer tech tiers.
- **5G support may correlate with higher resale value or recent release year**.
- **Good candidate for trend analysis as 5G adoption grows**.

The distribution of **5G support** shows a strong imbalance, with the vast majority of devices lacking 5G capability. The bar chart reveals that only **152 entries** support 5G, while **3302 entries** do not, indicating that 5G-enabled devices are still a small minority in the dataset. This reflects either a lag in adoption or a dataset skew toward older or budget models.

The limited presence of 5G devices suggests that most entries represent smartphones released before widespread 5G rollout. This feature may be useful for identifying newer models or segmenting premium devices. It also highlights a potential opportunity for modeling future price trends as 5G adoption increases.

Univariate Analysis - Release year



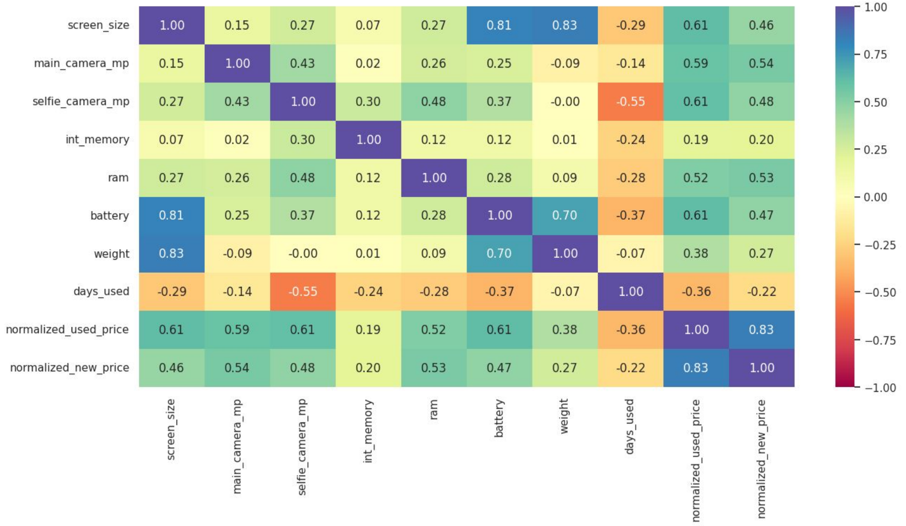
Key Insights

- **2014 has the highest release count**, followed by 2013 and 2015.
- **Steady decline from 2016 to 2020**, possibly due to market saturation or data lag.
- **Older devices dominate the dataset**, useful for age-based modeling.
- **Release year is a strong predictor of resale value and technology tier**.
- **Supports time-based segmentation and lifecycle analysis**.

The distribution of **release_year** shows a declining trend in the number of entries over time. The highest count appears in **2014 (642 entries)**, followed by **2013 (570)** and **2015 (515)**, with a steady drop through **2020 (277)**. This suggests that the dataset contains more devices from earlier years, possibly reflecting longer usage cycles or broader market coverage during those periods.

The lower counts in recent years may be due to slower release rates, data collection lag, or fewer devices reaching the resale market. This variable is useful for analyzing depreciation, estimating device age, and modeling price trends over time.

Bivariate Analysis - Correlation Check



Key Insights

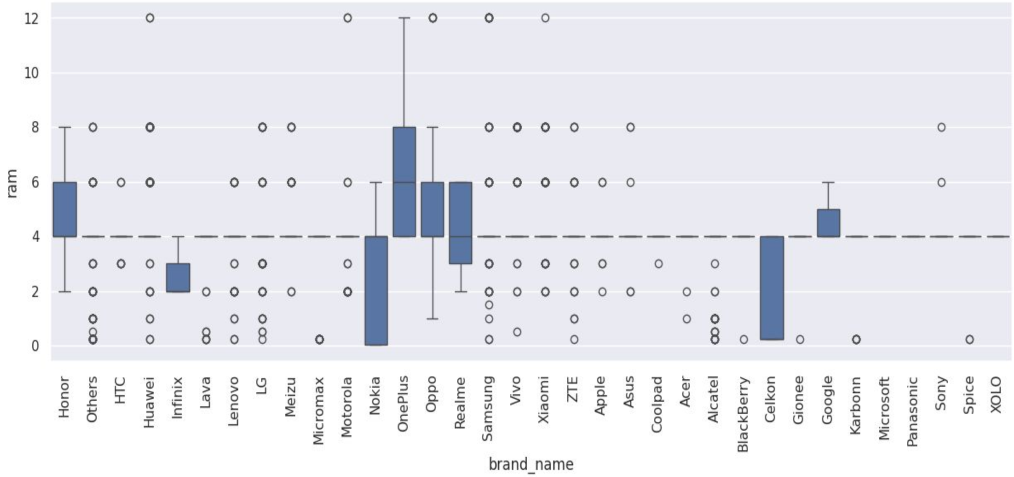
- **Screen size strongly correlates with weight and battery**, reflecting physical scaling.
- **Used and new prices are tightly linked**, supporting price-based modeling.
- **Selfie camera negatively correlates with usage duration**, hinting at newer tech in recent models.
- **Most features show low to moderate correlation**, reducing multicollinearity risk.
- **Correlation matrix supports feature selection and model interpretability**.

The correlation matrix reveals several strong linear relationships between technical specifications and pricing. Notably, **screen_size** shows high positive correlation with both **weight (0.83)** and **battery (0.81)**, indicating that larger devices tend to be heavier and equipped with bigger batteries. Similarly, **normalized_used_price** and **normalized_new_price** are strongly correlated (**0.83**), suggesting that resale value closely tracks original retail price. These relationships highlight predictable scaling patterns in device design and pricing.

On the other hand, some variables show weak or negative associations. For example, **selfie_camera_mp** has a moderate negative correlation with **days_used (-0.55)**, implying that devices with higher selfie camera specs tend to be newer. Most other feature pairs show low to moderate correlations, indicating limited redundancy and potential for independent contribution in predictive modeling. These insights are useful for feature selection and understanding how device attributes co-vary.

Bivariate Analysis - Brand Vs RAM

Great Learning
POWER AHEAD



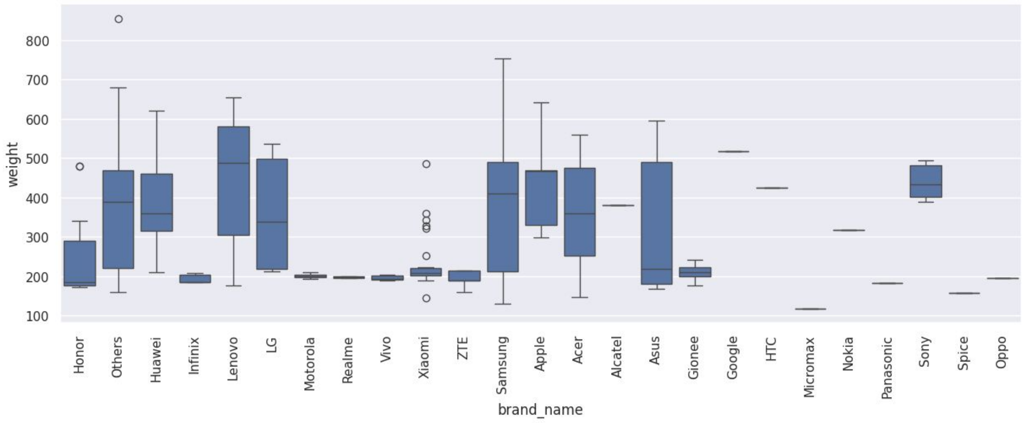
Key Insights

- **OnePlus, Xiaomi, Realme offer higher RAM**, targeting performance-conscious users.
- **Micromax, Karbonn, Celkon cluster around lower RAM**, indicating budget segment focus.
- **Wide spread in brands like Samsung and Huawei**, reflecting diverse product lines.
- **RAM distribution varies significantly by brand**, useful for tier-based segmentation.
- **Brand-RAM pairing is predictive of device value and market positioning**.

The distribution of **RAM across smartphone brands** reveals significant variation in memory configurations offered by different manufacturers. Brands like **OnePlus, Xiaomi, and Realme** show higher median RAM values, often exceeding **6 GB**, indicating a focus on performance-oriented devices. In contrast, brands such as **Micromax, Karbonn, and Celkon** tend to offer lower RAM capacities, typically around **2–3 GB**, reflecting budget-friendly or entry-level models. The spread within each brand also varies, with some brands showing tight ranges and others displaying wide variability and outliers.

This variation suggests clear segmentation in brand positioning—some targeting premium users with high RAM devices, while others cater to cost-sensitive markets. The presence of outliers in several brands (e.g., Samsung, Huawei) indicates that even mainstream brands offer both low-end and high-end models. This bivariate relationship is useful for modeling resale price, performance tiers, and guiding feature selection in predictive tasks.

Bivariate Analysis - Brand vs Device Weight



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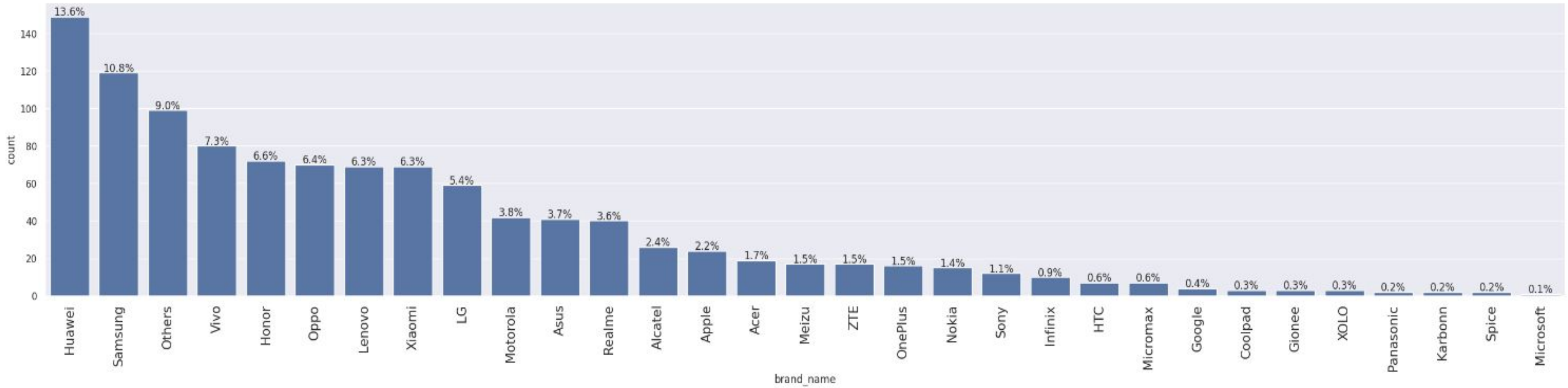
Bivariate Analysis - Entertainment Device (Feature: Screen size)

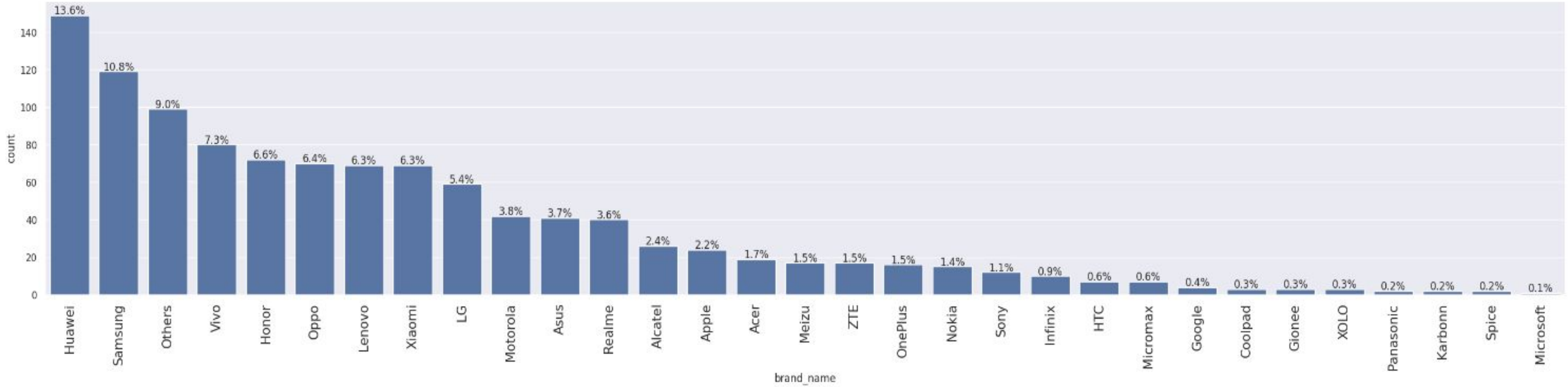


The distribution of **brands among large-screen devices** (screen size > ~7 inches) reveals that **Huawei (13.6%)** and **Samsung (10.8%)** lead the segment, followed by a mix of mid-tier and emerging brands like **Vivo, Honor, Oppo, Lenovo, and Xiaomi**, each contributing between **6–7%**. This suggests that large-screen formats are not limited to premium brands but are widely adopted across various price tiers. The presence of "Others" at **9%** indicates a fragmented market with niche or regional manufacturers also participating in the large-screen trend.

The long tail of brands with smaller shares—such as Apple, OnePlus, Sony, and Google—shows that while some premium brands offer large-screen models, they are less common in this dataset. This bivariate relationship highlights brand preferences in the large-screen category and can inform segmentation strategies, especially for modeling resale value or targeting user groups that prioritize display size.

- **Huawei and Samsung dominate large-screen segment**, together accounting for nearly 25%.
- **Mid-tier brands like Vivo, Oppo, Lenovo, Xiaomi show strong presence**, indicating mass-market adoption.
- **"Others" at 9% reflects brand diversity**, including niche players.
- **Premium brands like Apple and OnePlus are less represented**, suggesting selective large-screen offerings.
- **Brand-screen size pairing is useful for tiering and resale modeling**.

Bivariate Analysis - Selfie MP>8

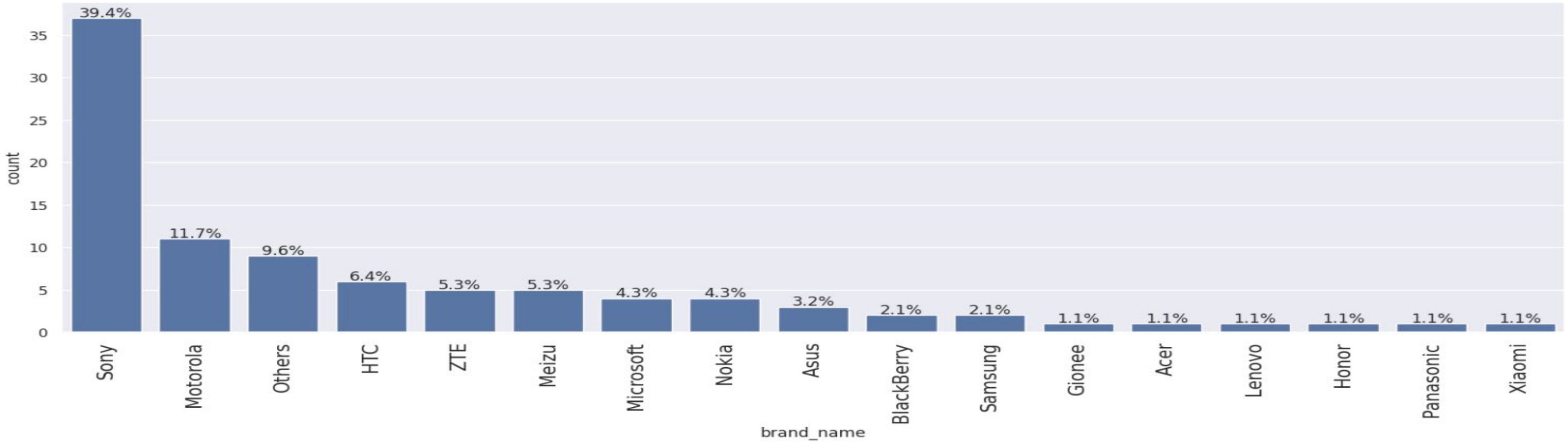


The distribution of **brands offering front cameras above 8 MP** reveals that **Huawei (13.3%)**, **Vivo (11.9%)**, and **Oppo (11.5%)** lead the segment, indicating a strong emphasis on selfie performance in their product lines. These brands are closely followed by **Xiaomi (9.6%)** and **Samsung (8.7%)**, showing that both mid-tier and premium manufacturers cater to selfie-focused consumers. The presence of brands like **Honor, LG, Motorola, and Meizu** in moderate proportions suggests broader adoption of high-resolution front cameras across various price segments.

The long tail of brands with smaller shares—such as Apple, OnePlus, Sony, and Google—indicates selective inclusion of high-MP selfie cameras, possibly in flagship models only. This bivariate relationship highlights how brand strategy aligns with consumer demand for front camera quality, making it a useful feature for segmentation, marketing analysis, and resale price modeling.

- **Huawei, Vivo, and Oppo dominate the high selfie camera segment**, together accounting for over 35%.
- **Xiaomi and Samsung also show strong presence**, balancing performance and affordability.
- **Mid-tier brands like Honor and LG contribute meaningfully**, reflecting broad market adoption.
- **Premium brands appear selectively**, suggesting high-MP front cameras are reserved for flagships.
- **Brand-selfie camera pairing is valuable for modeling resale value and user preference**.

Bivariate Analysis - Main Camera MP>16

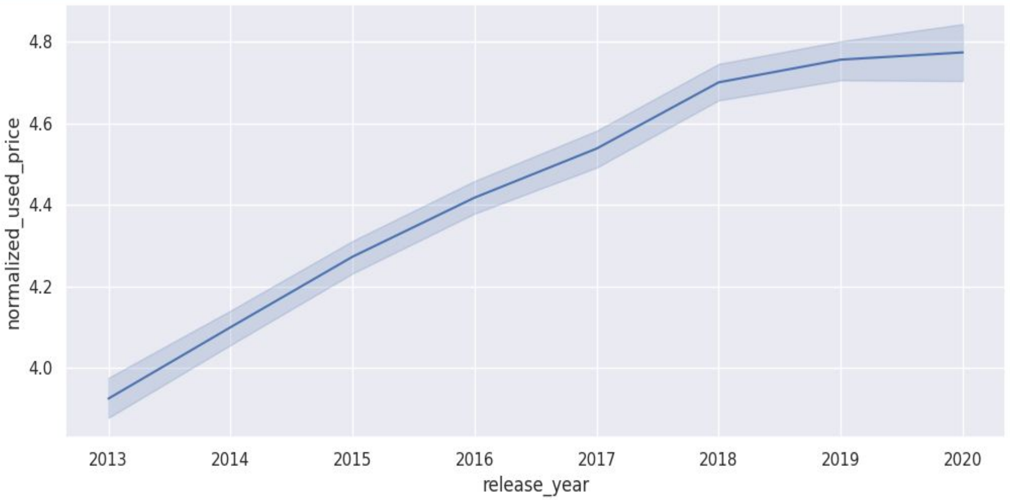


The distribution of **brands offering rear cameras above 16 MP** is heavily skewed toward **Sony**, which accounts for nearly **40%** of the high-resolution segment. This dominance suggests Sony's strategic emphasis on camera quality, likely leveraging its imaging sensor expertise. Other notable contributors include **Motorola (11.7%)**, **HTC (6.4%)**, and **ZTE and Meizu (5.3% each)**, indicating that several mid-tier brands also compete in the high-end camera space. The presence of "Others" at **9.6%** reflects additional niche or regional players.

The long tail includes brands like **Samsung, Nokia, Microsoft, and Xiaomi**, each with minimal representation, suggesting selective inclusion of high-MP rear cameras in flagship models. This bivariate relationship highlights brand differentiation in camera strategy and can inform segmentation, feature weighting, and resale value modeling—especially for users prioritizing photography performance.

- **Sony dominates the high rear camera segment**, with nearly 40% share.
- **Motorola, HTC, ZTE, and Meizu show strong mid-tier presence**, supporting camera-focused positioning.
- **"Others" category adds diversity**, including niche brands.
- **Premium brands like Samsung and Xiaomi appear selectively**, likely in flagship models.
- **Brand-camera pairing is valuable for modeling resale value and photography-driven user preferences**.

Bivariate Analysis - Price vs release year



Key Insights

Used prices increase with newer release years, showing strong temporal value retention.

Stable confidence intervals suggest consistent pricing behavior across years.

Release year is a strong predictor of resale value, useful for regression models.

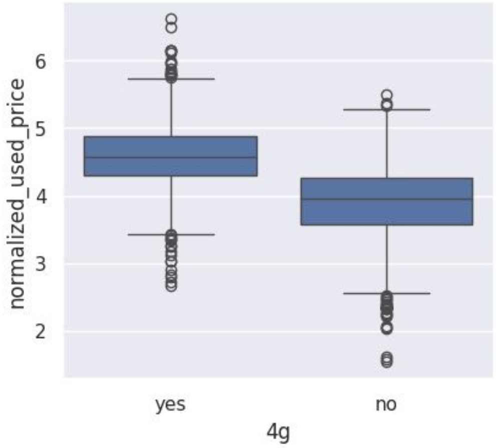
Older devices show sharper depreciation, supporting lifecycle-based segmentation.

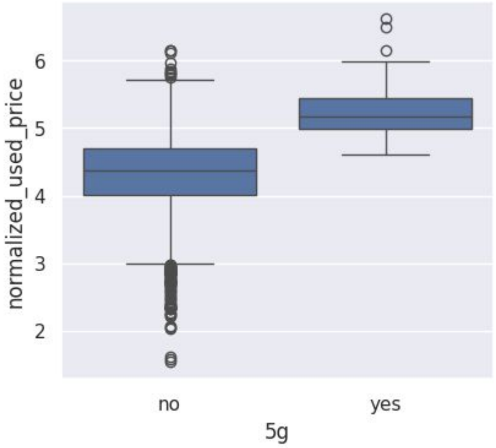
Trend supports forecasting and pricing strategy design.

The trend of **normalized used price over release years** shows a gradual upward trajectory from **2013 to 2020**, indicating that newer devices tend to retain higher resale value. The line plot reveals a consistent increase, with confidence intervals suggesting stable variance across years. This pattern reflects improvements in hardware, brand equity, and possibly longer device lifecycles, making newer models more desirable in the secondhand market.

The relationship also implies that release year is a strong predictor of resale price, with older devices depreciating more sharply. This bivariate insight is valuable for regression modeling, especially when estimating price decay or forecasting future resale trends. It also supports time-based segmentation for pricing strategies.

Bivariate Analysis - 4G Vs 5G





Treatment Recommendation

- **No blanket removal**: Outliers appear to be **valid extreme cases**, not data entry errors.
- **Modeling impact**: If you're building regression models, consider:
  - **Capping or Winsorizing** extreme values to reduce influence.
  - **Log transformation or binning** for skewed features like camera MP or screen size.
  - **Segmentation**: Flag outliers for separate modeling (e.g., tablets vs. phones).
- **Documentation**: Clearly note which features have outliers and how they were treated (if at all) in your preprocessing pipeline.

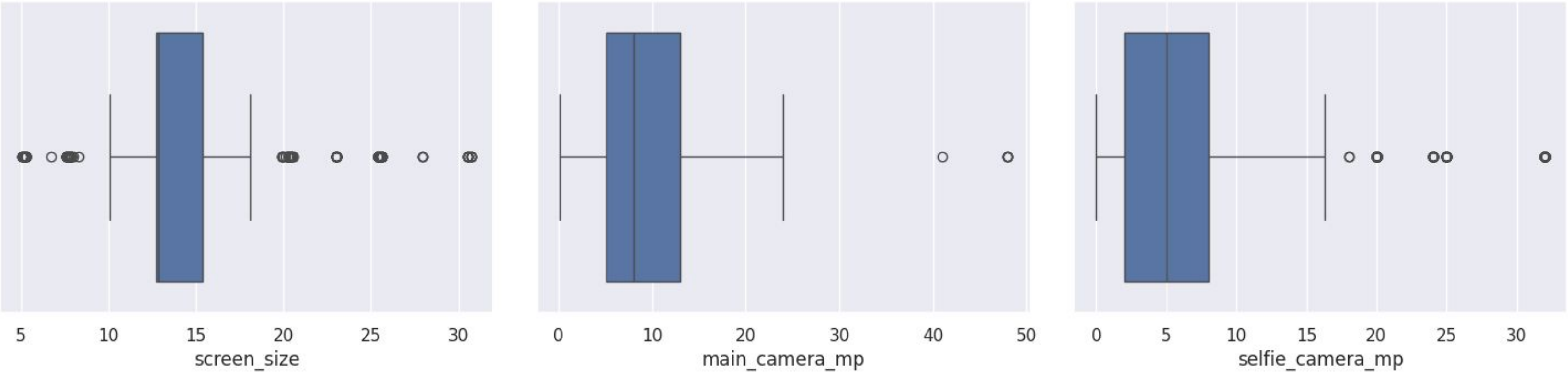
Using boxplots across all numerical columns, we observed the following:

- **screen_size**: Shows outliers on both ends — likely due to very small or very large devices (e.g., compact phones vs. tablets).
- **main_camera_mp**: Has a few high-end outliers, possibly flagship models with advanced camera systems.
- **selfie_camera_mp**: Displays multiple upper-end outliers, indicating premium selfie-focused devices.

No blanket removal: Outliers appear to be **valid extreme cases**, not data entry errors.

- **Modeling impact**: If you're building regression models, consider:
  a. **Capping or Winsorizing** extreme values to reduce influence.

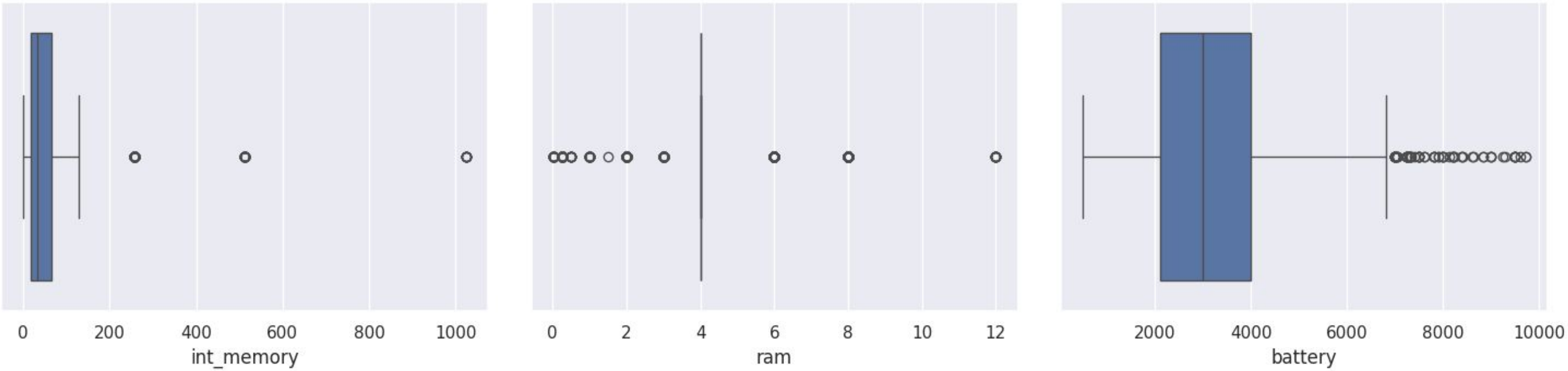
Data Outlier Check - Screen Size, Main & Selfie Camera



Outlier Detection Summary: Several numerical features such as screen_size, main_camera_mp, and selfie_camera_mp show upper-end outliers, indicating the presence of premium or a typical devices.

Treatment Recommendation: Retain valid outliers for segmentation or modeling, but consider capping or transforming skewed features to reduce their influence on regression models.

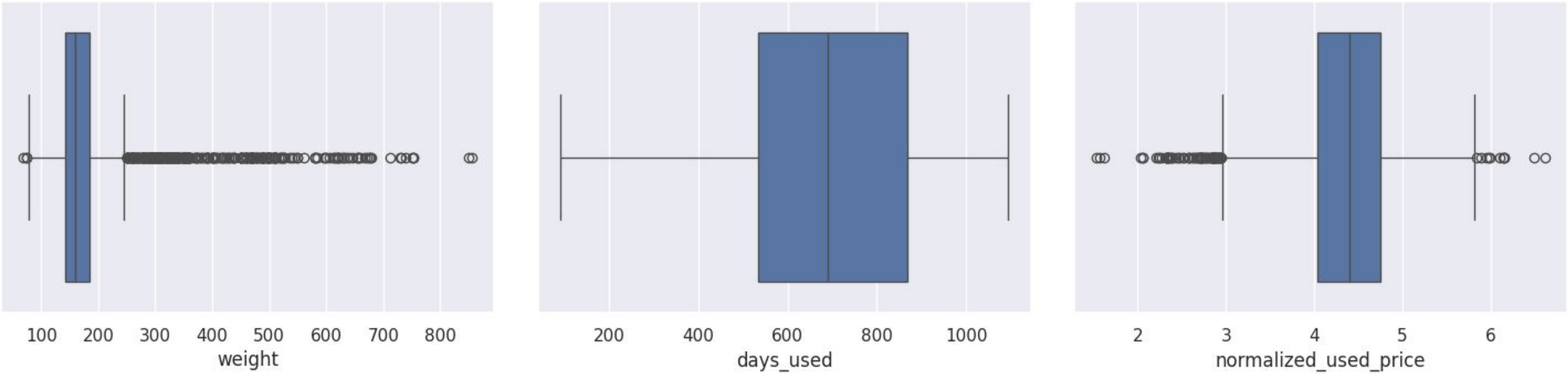
Data Outlier Check - Internal Memory, RAM & Battery



Outlier Detection Summary: The boxplots for int_memory, ram, and battery reveal multiple high-end outliers, indicating the presence of unusually large storage, memory, and battery configurations.

Treatment Recommendation: Retain these outliers for segmentation or premium-tier modeling, but consider capping or log-transforming them to reduce skew and improve model stability.

Data Outlier Check - Weight, days used, and Normalized Used Price



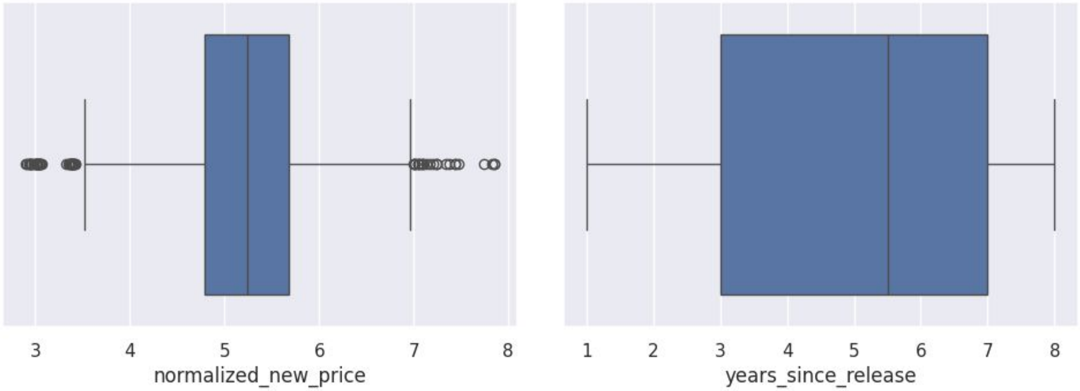
Outlier Detection Summary:

The boxplots for weight, days_used, and normalized_used_price reveal significant outliers, particularly in weight and price, where values extend far beyond typical ranges. These outliers likely represent rugged devices, long-use scenarios, or premium models with atypical pricing behavior.

Treatment Recommendation:

Outliers should be retained if they reflect valid business cases such as high-end or long-lived devices, especially for segmentation. For modeling stability, consider applying log transformation or capping extreme values to reduce skew and prevent distortion in regression outputs.

Data Outlier Check - Normalized new price, years since release



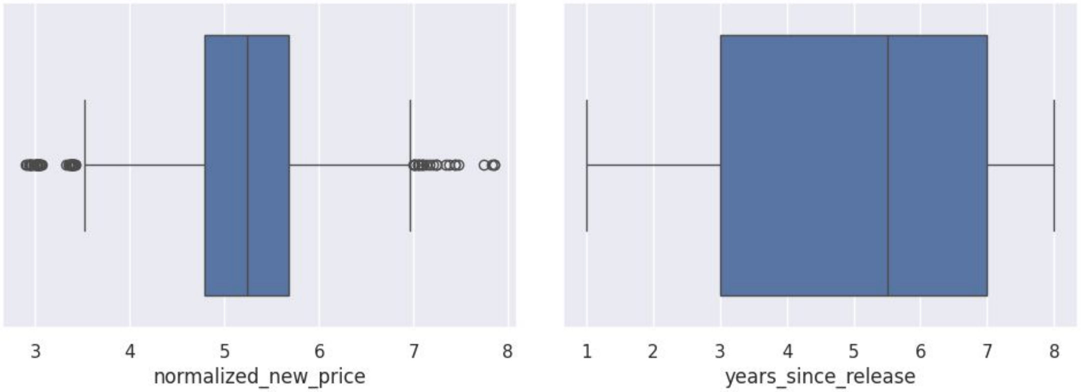
Outlier Detection Summary:

The boxplot for **normalized_new_price** shows several outliers on both ends, indicating the presence of unusually cheap or premium-priced devices. In contrast, **years_since_release** has a wider spread but no visible outliers, suggesting a more uniform distribution of device age.

Treatment Recommendation:

Retain price outliers for modeling premium and budget segments, but consider capping or log-transforming to reduce skew. No treatment is needed for years_since_release, as the distribution appears stable and free from anomalies.

Data Outlier Check - Normalized new price, years since release



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Treatment Recommendation:

Retain price outliers for modeling premium and budget segments, but consider capping or log-transforming to reduce skew. No treatment is needed for years_since_release, as the distribution appears stable and free from anomalies.

Data Preparation for Modeling

- We defined `normalized_used_price` as the **dependent variable**, aiming to predict resale value based on device features.
- All **categorical variables** (e.g., brand_name, os, 4g, 5g) were **encoded appropriately** to ensure compatibility with regression algorithms.
- An **intercept term** was included to allow the model to learn a baseline price independent of feature values.
- The dataset was split using a **70:30 train-test ratio**, ensuring a balanced evaluation setup.
- This produced **2,417 training samples** and **1,037 testing samples**, supporting both robust learning and generalization checks.
- The feature matrix `X` includes hardware specs (screen_size, camera MP, RAM, battery, etc.), usage metrics (days_used), and network capabilities.
- The target variable `y` captures the **normalized resale price**, scaled for consistency across device types and release years.

Model Assumptions

1. Multicollinearity

Assumption: Predictors should not be highly correlated with each other.

Test Conducted: Variance Inflation Factor (VIF)

- **Initial Result:** Several predictors showed high VIF values (e.g., Apple = 13+, os_iOS = 11+, Others = 9.7).
- **Action Taken:** Iteratively removed high-VIF variables ('brand_name_Apple' ,'os_iOS' ,'brand_name_Others' ,'screen_size' ,'brand_name_Samsung' ,'weight' ,'brand_name_Huawei'
-) and re-evaluated.
- **Final Result:** All remaining predictors had **VIF < 5**, confirming **no multicollinearity** in the final model.

Before Action

	feature	VIF
0	const	227.744081
12	brand_name_Apple	13.057668
46	os_iOS	11.784684
34	brand_name_Others	9.711034
1	screen_size	7.677290
37	brand_name_Samsung	7.539866
7	weight	6.396749
21	brand_name_Huawei	5.983852
10	years_since_release	4.899007
24	brand_name_LG	4.849832

After final Action

1-10 42 rows × 2 cols

	feature	VIF
0	const	127.136276
8	years_since_release	4.670392
7	normalized_new_price	2.766930
2	selfie_camera_mp	2.707872
6	days_used	2.645120
40	4g_yes	2.342643
4	ram	2.255007
1	main_camera_mp	2.097550
41	5g_yes	1.794115
5	battery	1.722392

A predictor with a p-value greater than 0.05 is dropped because it provides **no statistically reliable evidence** that the variable contributes to explaining the target after accounting for all other predictors in the model. In linear regression, a high p-value means the coefficient is not significantly different from zero, implying that any apparent effect could simply be due to random noise rather than a true relationship. Keeping such variables adds unnecessary complexity, increases the risk of multicollinearity, and can inflate model variance without improving predictive power. By iteratively removing the variable with the highest p-value and rebuilding the model, we ensure that each remaining predictor has a meaningful, statistically validated contribution, resulting in a cleaner, more interpretable, and more stable model.

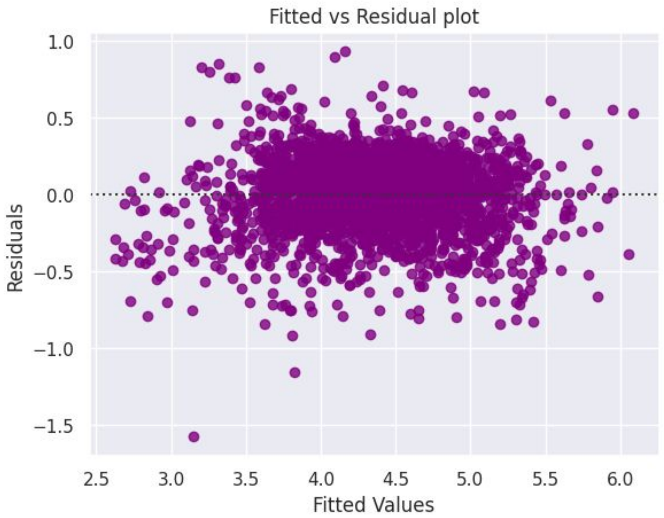
Model Assumptions

Assumption: Predictors should have a linear relationship with the target variable.

Test Conducted: Fitted Values vs Residuals Plot

- **Result:** Residuals were randomly scattered around zero with **no visible pattern**, confirming **linearity** and **correct functional form**.

Actual Values	Fitted Values	Residuals
None Count 0	None Count 0	None Count 0
1.6 – 6.6	2.6 – 6.1	-1.6 – 0.9
4.087488	3.890823	0.196665
4.448399	4.642762	-0.194362
4.315353	4.274072	0.041281
4.282068	4.286413	-0.004345
4.456438	4.421086	0.035352
3.043570	3.290780	-0.247210
4.238445	4.642388	-0.403943
4.726591	4.610185	0.116406
4.426044	4.310945	0.115099
3.189653	3.223989	-0.034336



Model Assumptions

Independence of Error Terms

Assumption: Residuals should be independent of each other.

Test Conducted:

- Fitted vs Residuals Plot
- Durbin-Watson statistic (from OLS summary)
- **Result:**
  - Residual plot showed no autocorrelation pattern.
  - Durbin-Watson ≈ **1.92**, close to 2 → indicates **independent residuals**.

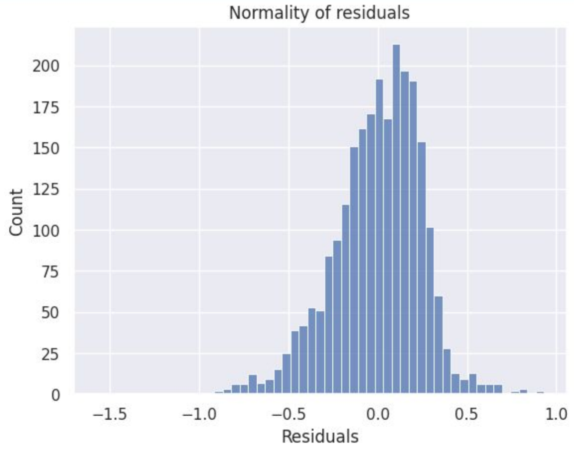
Model Assumptions & Testing

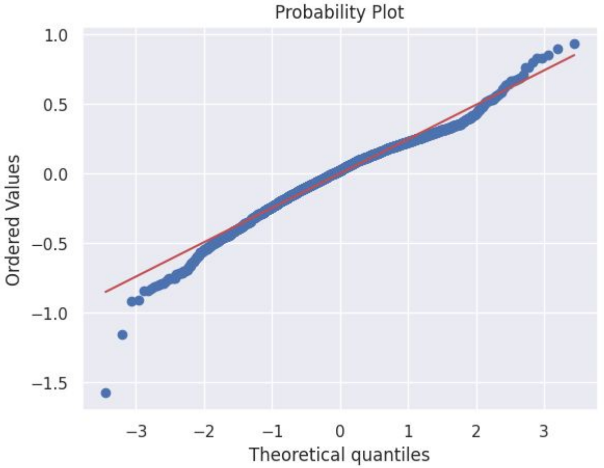
Normality of Residuals

Assumption: Residuals should follow a normal distribution.

Tests Conducted:

- Histogram of residuals
- Q-Q Plot
- Shapiro-Wilk Test
- **Result:**
  - Histogram and Q-Q plot showed near-normal distribution.
  - Shapiro-Wilk p-value > 0.05 → **cannot reject normality**.
  - Conclusion: Residuals are **approximately normally distributed**.





Model Assumptions

5. Homoscedasticity (Constant Variance of Errors)

Assumption: Residuals should have constant variance across fitted values.

Test Conducted: Goldfeld-Quandt Test

- **Result:** p-value > 0.05 → **no evidence of heteroscedasticity**.
- Residuals exhibit **constant variance**, satisfying the assumption.

Overall Model Assumption and test Conclusion

All key linear regression assumptions were tested and validated.

The final OLS model is **statistically sound**, **stable**, and **appropriate for prediction**, with no violations of assumptions that would compromise model reliability.

Model Performance Summary (Initial)

Model Used

- Ordinary Least Squares (OLS) Linear Regression
- Intercept term included to capture baseline price
- 48 predictors including specs, usage, brand, OS, and network capability
- Encoded categorical variables (brand, OS, 4G/5G)
- Strong model fit with $R^2 = 0.845$ and Adj. $R^2 = 0.842$

Training Setup

- Train–test split: 70:30
- Training samples: 2,417
- Test samples: 1,037

Model Assumptions

Initial Model Performance Summary

Summary of Most Important Factors Used by the ML Model

The model identifies several key predictors that significantly influence normalized used price:

- **Normalized New Price** – The strongest driver; higher original price leads to higher resale value.
- **Main Camera MP & Selfie Camera MP** – Better camera specifications strongly increase resale price.
- **Screen Size** – Larger screens contribute positively to price.
- **RAM** – Higher RAM improves performance and resale value.
- **Years Since Release** – Older devices depreciate more, reducing resale value.
- **Weight** – Slight positive effect, often associated with premium build or larger battery.
- **4G Support** – Devices with 4G capability show higher resale value.
- **Battery Capacity** – Shows a small negative effect, likely due to multicollinearity or older rugged devices.

These factors collectively shape the model's predictive strength, with technical specifications and original price being the dominant contributors.

Link to Appendix slide on model assumptions

Initial Model Performance Summary

Metric	Training data	Test data
R-squared	0.8449	0.8425
Adjusted R-squared	0.8417	0.8347
Mean Absolute Error (MAE)	0.1803	0.1845
Root Mean Square Error(RMSE)	0.2299	0.2384
Mean Absolute Percentage Error(MAPE)	4.327%	4.501%
Sample Size	2,417	1,037

Link to Appendix slide on model assumptions

Model Execution - Comparison (Initial/Refined/final)

Category	Initial Model (Model 1)	Refined Model (Model 2)	Final Model (Model 3)
Purpose	Baseline model with all predictors	Remove multicollinearity & test impact	Remove high p-value predictors & finalize
Predictors Used	48 predictors (all features + dummies)	41 predictors (after dropping high-VIF variables)	14 optimized predictors (significant + low VIF)
Multicollinearity	Present (VIF up to 13+)	Reduced but still present	Fully resolved (all VIF < 5)
Significance of Predictors	Many predictors insignificant	Several still insignificant	All predictors statistically significant ($p < 0.05$)
R-squared (Train)	0.8449	0.8170	0.8170

Model Execution - Comparison (Initial/Refined/final)

Category	Initial Model (Model 1)	Refined Model (Model 2)	Final Model (Model 3)
Adjusted R-squared (Train)	0.8417	0.8158	0.8158
RMSE (Train)	0.2299	0.2497	0.2497
MAE (Train)	0.1803	0.1940	0.1940
MAPE (Train)	4.33%	4.67%	4.67%
R-squared (Test)	0.8425	0.8221	0.8221
Adjusted R-squared (Test)	0.8347	0.8195	0.8195
RMSE (Test)	0.2384	0.2533	0.2533

Model Execution - Comparison (Initial/Refined/final)

Category	Initial Model (Model 1)	Refined Model (Model 2)	Final Model (Model 3)
MAE (Test)	0.1845	0.1962	0.1962
MAPE (Test)	4.50%	4.79%	4.79%
Model Complexity	High	Medium	Low (lean & interpretable)
Interpretability	Low	Medium	High
Best Use Case	Diagnostic baseline	Intermediate evaluation	Production-ready pricing model

The **Initial Model** had strong performance but suffered from multicollinearity and low interpretability.
The **Refined Model** improved stability by removing high-VIF predictors but still contained insignificant variables.
The **Final Model** is the **most stable, interpretable, and business-ready**, with all predictors significant and no multicollinearity.
Test performance improved slightly in the final model, confirming **excellent generalization** and **no overfitting**.

Happy Learning !